

# JINGPEI ZHONG

## EDUCATION

---

**University of Massachusetts Amherst**

*September 2022 - June 2026*

B.S. in Physics

Minor in Math

**Course work:** Computational Physics, Multivariate Calculus, Ordinary Differential Equations, Tech Theoretical Physics, Linear Algebra, Classical Mechanics, Honors Classical Mechanics, Quantum Mechanics, Statistical Mechanics, Honors Statistical Mechanics, Electricity and Magnetism, Fundamental Concepts of Math

**In-Progress Courses:** Mathematical Modeling, Quantum Computation, Applied Linear Algebra, Mathematics for Machine Learning, Stochastic Processes

## ACADEMIC RESEARCH EXPERIENCE

---

**Undergraduate Researcher, Department of Physics, UMass Amherst**

*September 2024 - May 2026*

Supervisor: Professor Jonathan Machta

- Implemented large-scale simulations of kinetic Ising models on HPC clusters using C++ and Python; investigated sequential versus parallel updating rules.
- Reproduced numerical results from core literature to validate model behavior, tracking magnetization susceptibility and heat capacity under different update schemes.
- Through simulations of spin systems, developed a solid understanding of stochastic processes and Monte Carlo simulations.
- Compared algorithmic performance between synchronous and asynchronous updating, analyzing the discrepancies in dynamic complexity.
- Explored a random memory update scheme in spin dynamics and analyzed its specific impact on critical point systems.

## INTERNSHIP EXPERIENCE

---

**Jiangsu Data-Win Cloud Network Technology Co. Ltd.**

*June 2025 - August 2025*

**AI Agent System Intern**

- Fine-tuned pre-trained language models for AI-driven customer interactions, improving response quality in real-world scenarios based on backend feedback datasets.
- Redesigned and tested prompt templates, achieving a 7-10% increase in response accuracy across multiple datasets.
- Maintained and preprocessed datasets using Python and Pandas to ensure data cleanliness for training and evaluation.
- Collaborated with backend AI agent systems to analyze user behavior and segment value tiers, expanding the potential customer funnel.
- Achieved an approximate 20% increase in product repurchase rate through data-driven optimization.
- Conducted multi-turn prompt experiments, bridging theoretical modeling experience from physics research to practical AI engineering applications.

## SKILL

---

- **Programming:** Python (Data Analysis, Data simulation, Data Plot), C,(Data Analysis, Data simulation), MATLAB

- **Data Analysis:** Computational Modeling, Statistical Methods, Error Analysis, Stochastic Processes in Networks, Critical Phenomena in Complex Networks
- **Tool:** LaTeX, Github, Python, C, VS Code, Linux, Adobe, Unity Cluster High Performance Computing
- **Laboratory:** RLC Circuit Analysis, Particle Detection, Simulations of a Physics Model.
- **Language :** Mandarin, Cantonese, English

## AWARD

---

UMass Chancellor's Award

*September 2022 - June 2026*